# Visvam Rajesh

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Full-stack software engineer building ML-driven autonomous systems and scalable distributed applications.

## Skills

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|---|---|
| **ML/Robotics** | PyTorch, scikit-learn, ROS 2, Motion Planning, URDF, MuJoCo, Computer Vision |
| **Systems & Backend** | Linux, Docker, AWS (EC2), Git, CI/CD, MongoDB, REST APIs, GraphQL |
| **Languages** | Python, C, C++, Rust, Java, JavaScript, TypeScript |
| **Core Skills** | Algorithm Design, System Architecture, Real-time Systems, Performance Optimization |

## Education

2025 - 2028 **Carnegie Mellon University** BS Computer Science & Robotics
**Coursework:** Data Structures, Algorithms, Computer Systems, Computer Vision

## Experience

**Adaptive Property Solutions** May 2026 - Aug 2026
**AI/ML Engineer Intern** Built automated data validation pipeline to ensure correctness across client jobs and 700+ technicians. Audited data governance across SharePoint/Dropbox, reducing repeat visits by 70%. Used GraphQL APIs to automate Matterport 3D Scan data collection. Trained image classification models using Open CLIP for incoming data validation, automating dataset creation using Gemma.

**Foam Robotics Lab, CMU Robotics Institute** Feb 2026 - May 2026
**Researcher, Motion Planning Lead** Architecting simulation and planning stack for DexKit surgical arm (C++/Python, ROS2, URDF/MuJoCo). Designed constraint-aware motion planning (MoveIt, Risk-DTRRT) with collision avoidance for safe tissue retractor manipulation.

**Carnegie Mellon Racing** Sept 2025 - Present
**Driverless Path-Planning Developer** Engineered real-time (20Hz) trajectory planning algorithm for autonomous racecar navigation (C++/ROS2). Optimized cone interpolation in Unreal Engine 4, enabling sub-60s lap completion with 6x faster trajectory computation.

**ScottyLabs** Sept 2025 - Present
**Software Engineer** Built high-performance Rust scheduling engine (MongoDB) optimizing course sequences by major constraints and minimizing walking distance. Achieved 6x performance improvement (1200ms→200ms) through algorithm optimization and backend tuning; maintained reliability with 5k+ concurrent users during peak registration.

## Projects

**twin** (https://github.com/BarrelDev/twin) Apr 2026
Built local-first AI knowledge base with semantic retrieval pipeline and tool-using agent runtime. Implemented custom Markdown chunker in Rust (PyO3 FFI) for performance-critical text processing and RAG pipeline with LanceDB; designed extensible multi-LLM provider interface with privacy-preserving architecture. Automated builds using Github Actions.

**Poké-ViT** (https://github.com/BarrelDev/poke-vit) Jan 2026
Built Vision Transformer from scratch in PyTorch (CUDA, RTX 4060) on 40k images across 1000 Pokémon classes. Achieved 56% top-1 accuracy with augmentation; engineered end-to-end pipeline with web scraping and preprocessing.

**Password Manager** (https://github.com/BarrelDev/password-manager) July 2025
Engineered cross-platform encrypted password manager in Python with AES-based Fernet encryption, salted hashing, and OS keychain session caching. Designed secure single-file storage with auto-locking sessions, fuzzy search, CLI interface, and TUI.

## Publications

Rajesh, V. & Wu, C. An Extension of Pathfinding Algorithms for Randomly Determined Speeds. IEEE International Performance, Computing, and Communications Conference, 2024. (Link)